T1 (a) the revised problem is:

$$\text{maximize} \quad z = -2x_1 + 5x_2 + 13x_3.$$

$$\text{s.t.} \quad x_2 + 3x_3 \le 20$$

$$5x_1 + 4x_2 + 10x_3 \le 90$$

$$x_1, x_2, x_3 \ge 0.$$

$\therefore c^T = (-5, 5, 13^T)$, $A = \begin{pmatrix} -1 & 1 & 3 \\ 12 & 4 & 10 \end{pmatrix}$, $B^{-1} = \begin{pmatrix} 1 & 0 \\ -4 & 1 \end{pmatrix}$

$y^* = (5, 0)$.

$\bar{c}^T = (-2, 5, 13)$, $\bar{A} = \begin{pmatrix} 0 & 1 & 3 \\ 5 & 4 & 10 \end{pmatrix}$

$\therefore y^*\bar{A} - \bar{c}^T = (5, 0)\begin{pmatrix} 0 & 1 & 3 \\ 5 & 4 & 10 \end{pmatrix} - (-2, 5, 13)$

$= (0, 5, 15) - (-2, 5, 13) = (2, 0, 2)$

$B^{-1}\bar{A} = \begin{pmatrix} 1 & 0 \\ -4 & 1 \end{pmatrix}\begin{pmatrix} 0 & 1 & 3 \\ 5 & 4 & 10 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 3 \\ 5 & 0 & -2 \end{pmatrix}$

So the revised problem, the tableau is:

BV	Row	Coefficient						RHS
		z	x_1	x_2	x_3	x_4	x_5	
z	(0)	1	2	0	2	5	0	100
x_2	(1)	0	0	1	3	1	0	20
x_5	(2)	0	5	0	-2	4	1	10

So the solution is optimal.

(b) the revised problem is:

$$\text{maximize} \quad z = -5x_1 + 6x_2 + 13x_3$$

$$\text{s.t.} \quad -x_1 + 2x_2 + 3x_3 \le 20$$

$$12x_1 + 5x_2 + 10x_3 \le 90$$

$$x_1, x_2, x_3 \ge 0.$$

$\therefore \bar{A} = \begin{pmatrix} -1 & 2 & 3 \\ 12 & 5 & 10 \end{pmatrix}$, $\bar{c}^T = (-5, 6, 13)$.

$c_B^T = (5, 0)$, $B^{-1} = \begin{pmatrix} 1 & 0 \\ -4 & 1 \end{pmatrix}$, $y^* = (5, 0)$

$b = (20, 90)^T$ $\quad \therefore y^* b = (5, 0)\begin{pmatrix} 20 \\ 90 \end{pmatrix} = 100$

$B^{-1} b = \begin{pmatrix} 1 & 0 \\ -4 & 1 \end{pmatrix}\begin{pmatrix} 20 \\ 90 \end{pmatrix} = \begin{pmatrix} 20 \\ 10 \end{pmatrix}$, $y^* \bar{A} - \bar{c}^T = (5, 0)\begin{pmatrix} -1 & 2 & 3 \\ 12 & 5 & 10 \end{pmatrix} - (-5, 6, 13)$

$= (-5, 10, 15) - (-5, 6, 13) = (0, 4, 2)$

So the revised problem, the tableau is:

BV	Row	Coefficient						RHS
		z	x_1	x_2	x_3	x_4	x_5	
z	(0)	1	0	4	2	5	0	100
x_2	(1)	0	-1	2	3	1	0	20
x_5	(2)	0	12	-3	-2	-4	1	10

After Gaussian elimination:

BV	Row	Coefficient						RHS
		z	x_1	x_2	x_3	x_4	x_5	
z	(0)	1	2	0	-4	3	0	60
x_2	(1)	0	$-\frac{1}{2}$	1	$\frac{3}{2}$	$\frac{1}{2}$	0	10
x_5	(2)	0	$\frac{29}{2}$	0	$\frac{5}{2}$	$-\frac{5}{2}$	1	40

$\because -4 < 0 \Rightarrow$ the solution is feasible but not optimal.

(c) the original problem:

maximize $z = -5x_1 + 5x_2 + 13x_3 + 0 \cdot x_6$

s.t. $-x_1 + x_2 + 3x_3 + x_4 + 0 \cdot x_6 = 20$

$12x_1 + 4x_2 + 10x_3 + x_5 + 0 \cdot x_6 = 90$

$x_1, x_2, x_3, x_4, x_5, x_6 \geq 0.$

$\therefore c^T = (-5, 5, 13, 0)$, $A = \begin{pmatrix} -1 & 1 & 3 & 0 \\ 12 & 4 & 10 & 0 \end{pmatrix}$

$c_B^T = (5, 0)$, $B = \begin{pmatrix} 1 & 0 \\ 4 & 1 \end{pmatrix}$, $y^* = c_B^T B^{-1} = (5, 0)$

the revised problem:

maximize $z = -5x_1 + 5x_2 + 13x_3 + 10x_6.$

s.t. $-x_1 + x_2 + 3x_3 + 3x_6 + x_4 = 20$

$12x_1 + 4x_2 + 10x_3 + 5x_6 + x_5 = 90$

$x_1, \cdots, x_6 \geq 0.$

$\therefore \bar{c}^T = (-5, 5, 13, 10)$, $\bar{A} = \begin{pmatrix} -1 & 1 & 3 & 3 \\ 12 & 4 & 10 & 5 \end{pmatrix}$

$\therefore y^* \bar{A} - \bar{c}^T = (5, 0) \begin{pmatrix} -1 & 1 & 3 & 3 \\ 12 & 4 & 10 & 5 \end{pmatrix} - (-5, 5, 13, 10)$

$= (-5, 5, 15, 15) - (-5, 5, 13, 10) = (0, 0, 2, 5)$

$B^{-1}\bar{A} = \begin{pmatrix} 1 & 0 \\ -4 & 1 \end{pmatrix} \begin{pmatrix} -1 & 1 & 3 & 3 \\ 12 & 4 & 10 & 5 \end{pmatrix} = \begin{pmatrix} -1 & 1 & 3 & 3 \\ 16 & 0 & -2 & -7 \end{pmatrix}$

$\therefore$ the revised problem's tableau is:

BV	Row	Coefficient							RHS
		z	x_1	x_2	x_3	x_6	x_4	x_5	
z	(0)	1	0	0	2	5	5	0	100
x_2	(1)	0	-1	1	3	3	1	0	20
x_5	(2)	0	16	0	-2	-7	-4	1	10

$\therefore$ the solution is feasible and optimal.

(d) $\therefore c^T = (-5, 5, 13)$, $A = \begin{pmatrix} -1 & 1 & 3 \\ 12 & 4 & 10 \end{pmatrix}$

$c_B^T = (5, 0)$, $B = \begin{pmatrix} 1 & 0 \\ 4 & 1 \end{pmatrix}$, $y^* = c_B^T B^{-1} = (5, 0)$

$\therefore$ the BF solution is $(0, 20, 0, 0, 10)$

$\therefore 3 \times 20 \geq 50$

After introducing new constraint.

$2x_1 + 3x_2 + 5x_3 + x_6 = 50$

$\therefore \bar{b} = (20, 90, 50)^T$, $\bar{A} = \begin{pmatrix} -1 & 1 & 3 \\ 12 & 4 & 10 \\ 2 & 3 & 5 \end{pmatrix}$, $\bar{c}_B^T = (5, 0, 1)$

$\bar{B} = \begin{pmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 3 & 0 & 1 \end{pmatrix}$, $\bar{B}^{-1} = \begin{pmatrix} 1 & & \\ -4 & 1 & \\ -3 & & 1 \end{pmatrix}$, $c^T = (-5, 5, 13)$.

$\therefore y^* = \bar{c}_B^T \bar{B}^{-1} = (5, 0, 1) \begin{pmatrix} 1 & & \\ -4 & 1 & \\ -3 & & 1 \end{pmatrix} = (2, 0, 1)$

$y^* \bar{A} - \bar{c}^T = (2, 0, 1) \begin{pmatrix} -1 & 1 & 3 \\ 12 & 4 & 10 \\ 2 & 3 & 5 \end{pmatrix} - (-5, 5, 13) = (5, 0, -2)$

$\bar{B}^{-1} \bar{A} = \begin{pmatrix} 1 & & \\ -4 & 1 & \\ -3 & & 1 \end{pmatrix} \begin{pmatrix} -1 & 1 & 3 \\ 12 & 4 & 10 \\ 2 & 3 & 5 \end{pmatrix} = \begin{pmatrix} -1 & 1 & 3 \\ 16 & 0 & -2 \\ 5 & 0 & -4 \end{pmatrix}$, $y^* \bar{b} = 90$

$\bar{B}^{-1} \bar{b} = \begin{pmatrix} 1 & & \\ -4 & 1 & \\ -3 & & 1 \end{pmatrix} \begin{pmatrix} 20 \\ 90 \\ 50 \end{pmatrix} = \begin{pmatrix} 20 \\ 10 \\ -10 \end{pmatrix}$

So the revised problem tableau is:

BV	Row	Coefficient							RHS
		z	x_1	x_2	x_3	x_4	x_5	x_6	
z	(0)	1	5	0	-2	2	0	1	90
x_2	(1)	0	-1	1	3	1	0	0	20
x_5	(2)	0	16	0	-2	-4	1	0	10
x_6	(3)	0	5	0	-4	-3	0	1	-10

After Gaussian elimination:

BV	Row	Coefficient							RHS
		z	x_1	x_2	x_3	x_4	x_5	x_6	
z	(0)	1	0	0	2	5	0	0	100
x_2	(1)	0	-1	1	3	1	0	0	20
x_5	(2)	0	16	0	-2	-4	1	0	10
x_6	(3)	0	5	0	-4	-3	0	1	-10

$\because -10<0 \Rightarrow$ this problem is infeasible.

T2 According to the title:

$$A=\begin{pmatrix} 1 & 3 & 4 \\ 1 & 4 & -1 \end{pmatrix},\quad B=\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix},\quad B^{-1}=\begin{pmatrix} 0 & 1 \\ 1 & -1 \end{pmatrix}$$

$$c^T=(2,7,-3),\quad C_B^T=(2,0)$$

$$y^*=C_B^T\cdot B^{-1}=(0,2),\quad b^T=(30,10),\quad \bar{b}^T=(30+3\theta,10-\theta)$$

So the revised problem's tableau is:

BV	Row	Coefficient						RHS
		z	x_1	x_2	x_3	x_4	x_5	
z	(0)	1	0	1	1	0	2	$20-2\theta$

x_1	(1)	0	0	-1	5	1	-1	$10-\theta$
x_4	(2)	0	1	4	-1	0	1	$20+4\theta$

if the new solution is feasible.

$$\therefore \begin{cases} 10-\theta \geq 0 \\ 20+4\theta \geq 0 \end{cases} \Rightarrow \begin{cases} \theta \leq 10 \\ \theta \geq -5 \end{cases} \Rightarrow \theta \in [-5, 10]$$

T3 (a) the primal problem:

$$\begin{pmatrix} 1 & -6 & -8 \\ 0 & 5 & 2 \\ 0 & 1 & 2 \end{pmatrix} \begin{pmatrix} z \\ x_1 \\ x_2 \end{pmatrix} \leq \begin{pmatrix} 0 \\ 20 \\ 10 \end{pmatrix}$$

$$\therefore A = \begin{pmatrix} 5 & 2 \\ 1 & 2 \end{pmatrix} \quad \therefore A^T = \begin{pmatrix} 5 & 1 \\ 2 & 2 \end{pmatrix}$$

so the dual problem is:

minimize $z = 20y_1 + 10y_2$

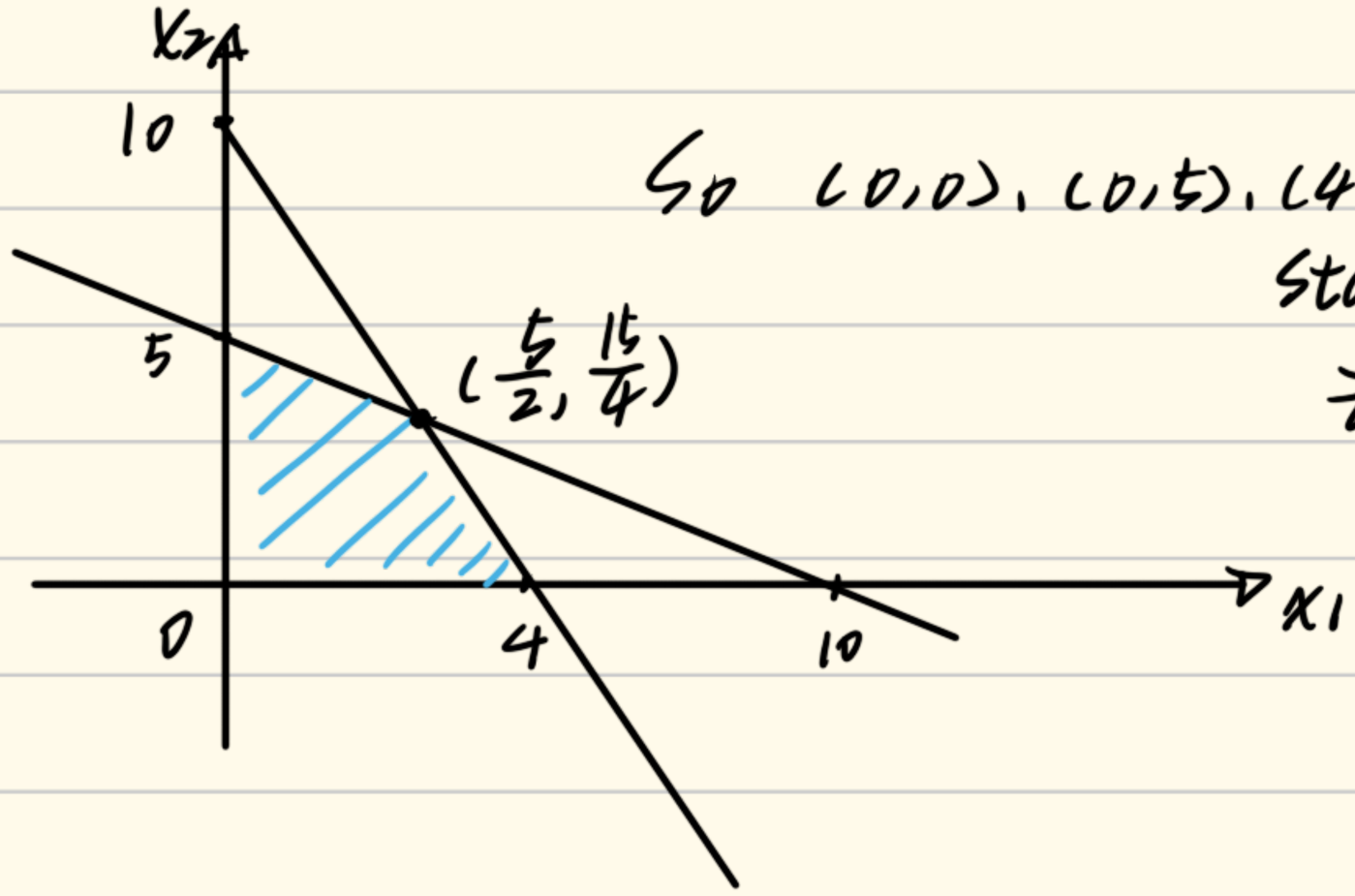
s.t. $\begin{pmatrix} 5 & 1 \\ 2 & 2 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \geq \begin{pmatrix} 6 \\ 8 \end{pmatrix}$

$$\Rightarrow \begin{cases} 5y_1 + y_2 \geq 6 \\ 2y_1 + 2y_2 \geq 8. \end{cases}$$

$y_1, y_2 \geq 0$.

(b) I. consider the primal problem:

$x_2 \leq 10 - 2.5x_1$; $x_2 \leq 5 - 0.5x_1$; $x_2 = -\frac{3}{4}x_1 + \frac{7}{8}$; $x_1 \geq 0$; $x_2 \geq 0$.



So $(0,0)$, $(0,5)$, $(4,0)$, $(\frac{5}{2}, \frac{15}{4})$ is CPF solution.

Start with $(0,0)$:

$z(0,0) = 0$;

$(0,5)$: $z(0,5) = 40$;

$(\frac{5}{2}, \frac{15}{4})$: $z(\frac{5}{2}, \frac{15}{4}) = 45$

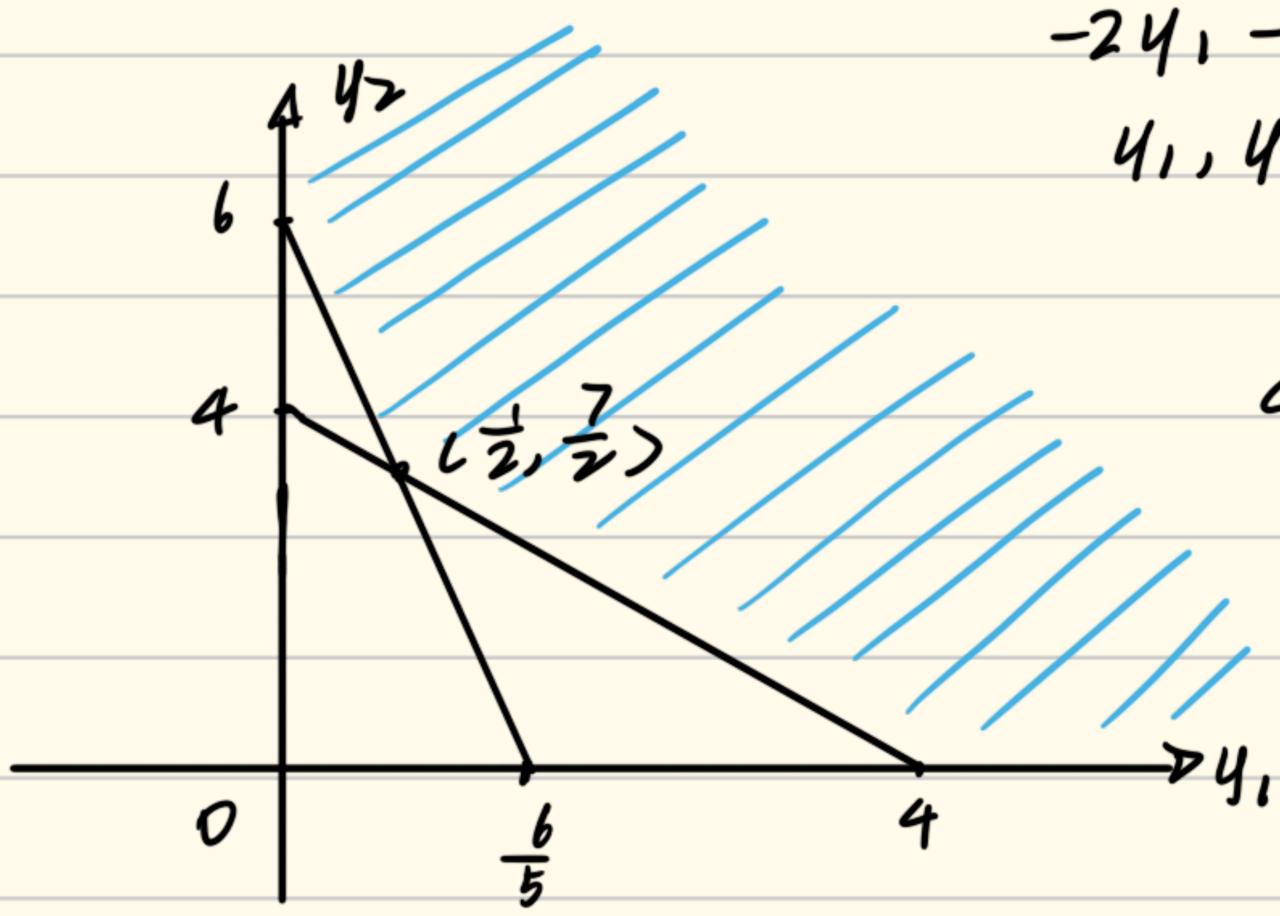
$(4,0)$: $z(4,0) = 24$

So the $(\frac{5}{2}, \frac{15}{4})$ is the optimal solution.
Otherwise, $(0,10)$、$(10,0)$ is the corner point infeasible solution.
$z(0,10)=80$; $z(10,0)=60$, $z(0,0)=0$.

II. consider the dual problem:

$$\begin{aligned} \text{minimize} \quad & z = 20y_1 + 10y_2 \\ \text{s.t} \quad & 5y_1 + y_2 \geq 6 \\ & 2y_1 + 2y_2 \geq 8 \\ & y_1, y_2 \geq 0 \end{aligned}$$

$$\Rightarrow \begin{aligned} \text{maximize} \quad & z = -20y_1 - 10y_2 \\ \text{s.t} \quad & -5y_1 - y_2 \leq -6 \\ & -2y_1 - 2y_2 \leq -8 \\ & y_1, y_2 \geq 0. \end{aligned}$$



So the blue section is feasible section.
$(\frac{1}{2}, \frac{7}{2})$、$(0,6)$、$(4,0)$ are the CPF solutions,
and $(0,4)$、$(0,0)$、$(\frac{6}{5},0)$ are the corner-point infeasible solution.

$\therefore z(0,6) = -60$; $z(\frac{1}{2}, \frac{7}{2}) = -45$; $z(4,0) = -80$
So $(\frac{1}{2}, \frac{7}{2})$ is the optimal solution.
$z(0,4) = -40$, $z(0,0) = 0$; $z(\frac{6}{5},0) = -24$.

(c) According to the title
the pairs of primal basic solution are:

$(0, 0, \underbrace{20, 10}_{BV})$、$(0, \underbrace{5, 10}_{BV}, 0)$, $(\underbrace{4, 0, 0, 6}_{BV})$

$(\overbrace{\frac{5}{2}, \frac{15}{4}}^{BV}, 0, 0)$、$(0, 10, 0, -10)$、$(10, 0, 30, 0)$

the pairs of complementary dual basic solution are:

$(0,0,-6,-8)$, $(0,4,-2,0)$, $(\frac{6}{5},0,0,\frac{-28}{5})$, $(\frac{1}{2},\frac{7}{2},0,0)$

$(0,6,0,4)$, $(4,0,14,0)$

$\because b^T=(20,10)$, $c^T=(6,8)$, $c^Tx=b^Ty$

$$\therefore \begin{cases} (0,0,20,10) \Leftrightarrow (0,0,-6,-8) \\ (0,5,10,0) \Leftrightarrow (0,4,-2,0) \\ (4,0,0,6) \Leftrightarrow (\frac{6}{5},0,0,\frac{-28}{5}) \\ (\frac{5}{2},\frac{15}{4},0,0) \Leftrightarrow (\frac{1}{2},\frac{7}{2},0,0) \\ (0,10,0,-10) \Leftrightarrow (0,6,0,4) \\ (10,0,-30,0) \Leftrightarrow (4,0,14,0) \end{cases}$$

The wavy lines represent BV.